Training-Free Token Reduction on MotionBench

Benchmark Results for Eleven Methods across Two Backbones

VLLM Project

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Abstract

This report presents benchmark results for eleven training-free video-LLM token-reduction methods on MotionBench (8,052 questions, 4,018 scoreable), evaluated on two backbones at a standardized 32 frames and 15% nominal token retention. Headline numbers: the Qwen3-VL-8B baseline reaches 62.52% against LLaVA-OV-7B's 52.66%, a 9.86-point gap that holds across all six question categories. On LLaVA-OV seven of nine methods are accuracy-neutral, falling inside the $\pm$1.54-point resolution floor, while two collapse to 36.78% and 38.20%. On Qwen3-VL, at identical settings, all ten portable methods lose accuracy, ranging from -0.35 to -6.55, and eight of those losses are statistically significant — so the ranking does not transfer between backbones. A controlled experiment on the two collapsed methods shows their loss does not depend on which tokens are kept: selecting by attention, at random, or evenly across all frames scores within 1.1 points, while the same code path retaining every token recovers the baseline. On this backbone a 15% budget costs roughly 15 points regardless of selection rule. Results are reported overall and by category; methodology, verification and limitations are in the appendices.

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1 Setup

Benchmark. MotionBench: 8,052 four-option multiple-choice questions over video clips. 4,034 carry an `NA` ground truth and are excluded, leaving **4,018 scoreable items** — the denominator for every accuracy figure below. Six question categories, of unequal size (Table 2). Scoring letter-matches A–D against ground truth; generation is greedy.

Backbones. **LLaVA-OV-7B** (SigLIP-so400m vision tower, Qwen2-7B LLM, mid-2024) and **Qwen3-VL-8B** (`Qwen3VLForConditionalGeneration`, a native architecture rather than a LLaVA derivative, late-2025).[1]

Configuration. All methods run at 32 frames. Methods exposing a single retention parameter are set to 15%; those whose budget is fixed internally, or that select frames rather than tokens, keep published defaults. Comparison is therefore at a common *nominal* setting, not at equal compute — measured budgets range from 7.5% to 50% (Appendix A).

Reading the numbers. At $n = 4{,}018$ the binomial 95% confidence half-width is ±1.54 points; smaller differences are not resolvable and are not ranked. * marks a difference significant by McNemar's test on paired predictions ($\chi^2 \geq 3.84$, $p < 0.05$); paired counts are in Appendix B. Every figure here was recomputed from per-sample prediction records (Appendix A).

2 Results

2.1 Summary

Table 1: All methods on both backbones, sorted by Qwen3-VL result. * = significant vs. that backbone's baseline. The dash marks a cell where the method is not implementable on that backbone.

	LLaVA-OV-7B		Qwen3-VL-8B	
Method	Overall	Δ	Overall	Δ
Baseline	*52.66*	—	*62.52*	—
PruneVID[†]	38.20	–14.46*	62.17	–0.35
DyCoke	53.36	+0.70	61.85	–0.67
STTM	51.72	–0.95	61.60	–0.92*
HoliTom	53.14	+0.47	60.33	–2.19*
MDP3	53.06	+0.40	59.66	–2.86*
FlashVID	53.31	+0.65	59.36	–3.16*
FastV	36.78	–15.88*	59.01	–3.51*
VisionZip (ctx.)[‡]	—	—	58.81	–3.71*
VideoITG	52.86	+0.20	56.35	–6.17*
AIM	52.86	+0.20	55.97	–6.55*

[1] Project records refer to a `qwen_1_5` template on LLaVA-OV. That is a prompt template, not a different LLM: `qwen_1_5` and `qwen_2` render byte-identical prompts here and the model is Qwen2-7B either way.

[†] PruneVID's only published operating point is `cluster_ratio=0.5` (50% retention), not 15% — it was never run at 15% on either backbone (Appendix A). [‡] Contextual half only — Qwen3-VL's vision tower has no CLS token, so VisionZip's dominant half is not implementable. Not the published method. VisionZip's complete form runs on LLaVA-1.5-7B (Table 11). DyTo appears in Table 11: it is bound to a Vicuna backbone and runs on neither of these two.

Table 1 contains the report's main observation. Reading across rows rather than down columns, the ordering does not survive the change of backbone: PruneVID is the worst result on LLaVA-OV and the best on Qwen3-VL; FlashVID is indistinguishable from the backbone on LLaVA-OV and loses 3.16 points on Qwen3-VL. Every Qwen3-VL delta is negative; six of nine LLaVA-OV deltas are positive but none resolvably so. A ranking measured on one backbone does not transfer to the other.

2.2 Backbones

Table 2: Baseline accuracy overall and by category (% correct within category). AO = Action Order, CM = Camera Motion, LM = Location-related Motion, MR = Motion Recognition, MO = Motion-related Objects, RC = Repetition Count. Chance is 25%.

Backbone	Overall	AO	CM	LM	MR	MO	RC
LLaVA-OV-7B	52.66	40.5	45.2	55.5	57.0	71.2	23.8
Qwen3-VL-8B	62.52	46.1	63.1	65.0	67.3	79.0	33.8
Δ	+9.86	+5.6	+17.9	+9.5	+10.3	+7.8	+10.0
n	4,018	519	385	546	1,478	690	400

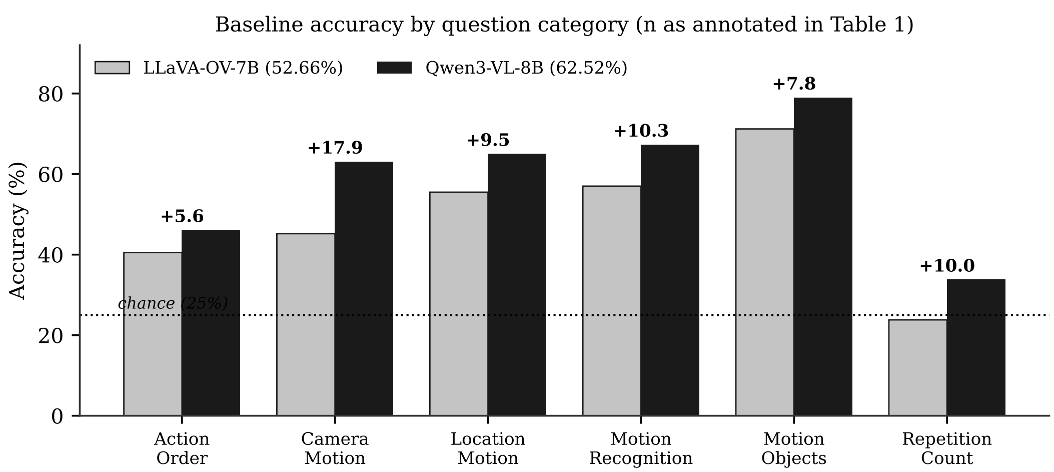

Figure 1: Baseline accuracy by category. Qwen3-VL-8B leads in all six, most sharply on Camera Motion (+17.9).

The 9.86-point backbone gap is larger than any method effect measured anywhere in this report, and it holds in every category.

Two properties of Table 2 govern how later tables should be read. Both backbones are strongest on Motion-related Objects (71.2%, 79.0%), the category most answerable from object appearance without integrating motion. And **Repetition Count is at or near chance for both** (23.8% and 33.8% against 25%): LLaVA-OV is not distinguishable from guessing there, so movement in that column is not evidence about a method and is not interpreted below.

2.3 LLaVA-OV-7B

Table 3: LLaVA-OV-7B, 32 frames, 15% nominal retention except where the method's budget is fixed internally. All nine methods that run on this backbone. Categories as in Table 2. VisionZip runs on a different backbone (Table 11). § AIM's released code ships two active merge steps, so it runs at 25% retention, not 15%. † PruneVID-OV runs at `cluster_ratio=0.5` (50% retention), the authors' only published point, never 15%.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
DyCoke	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.3
FlashVID	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM§	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.3
Baseline	*52.66*	—	*40.5*	*45.2*	*55.5*	*57.0*	*71.2*	*23.8*
STTM	51.72	–0.95	39.9	48.1	53.8	53.5	70.6	28.8
PruneVID-OV*†	38.20	–14.46	32.0	34.8	35.5	38.4	52.9	27.3
FastV*	36.78	–15.88	32.8	31.9	34.1	35.7	53.8	25.3

Seven of the nine methods land within 0.95 points of the backbone, inside the ±1.54-point floor, and none is significant (largest $\chi^2 = 2.34$ against a 3.84 threshold). They are statistically indistinguishable from the backbone and from each other; the ordering among them is not meaningful. Their category profiles track the backbone closely, with the largest deviations falling in the two smallest categories. At a 15% budget on this backbone, seven of nine token-reduction methods are accuracy-neutral.

FastV and PruneVID-OV are the exceptions, losing more than 14 points each, significantly, and degrading in every category carrying above-chance backbone signal. §2.8 establishes what separates them from the other seven, and §2.9 isolates the cause by experiment.

2.4 Qwen3-VL-8B

Table 4: Qwen3-VL-8B, 32 frames, 15% nominal retention except where noted. All ten portable methods. VisionZip is the contextual half only (see Table 1 notes). FlashVID and STTM run the authors' own code at published settings and no longer carry an off-spec caveat (Appendix C). † PruneVID runs at `cluster_ratio=0.5` (50% retention), the authors' only published point, never 15%.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
Baseline	*62.52*	—	*46.1*	*63.1*	*65.0*	*67.3*	*79.0*	*33.8*
PruneVID†	62.17	–0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	61.85	–0.67	45.1	62.6	63.9	66.4	78.4	34.5
STTM*	61.60	–0.92	45.9	61.6	64.8	66.0	77.1	34.8
HoliTom*	60.33	–2.19	44.7	61.3	61.0	66.4	76.2	29.0
MDP3*	59.66	–2.86	44.9	64.7	62.8	64.0	77.4	23.0
FlashVID*	59.36	–3.16	43.5	57.4	62.6	65.6	77.2	23.5
FastV*	59.01	–3.51	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (ctx.)*	58.81	–3.71	43.2	57.4	61.9	64.2	74.3	29.5
VideoITG*	56.35	–6.17	45.3	53.5	59.3	60.2	75.4	22.2
AIM*	55.97	–6.55	45.9	56.6	59.5	58.3	72.2	27.0

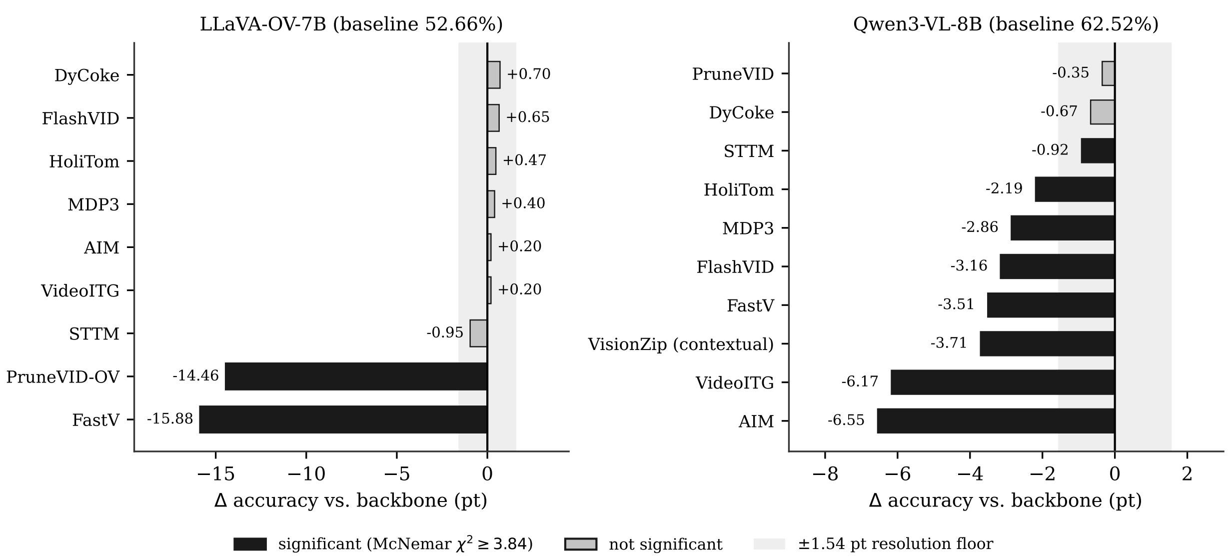

Figure 2: Change vs. backbone on each model. Shaded band is the ±1.54-point resolution floor. On LLaVA-OV every non-collapsed method falls inside it; on Qwen3-VL eight of ten fall outside it and all ten are negative.

Every method loses accuracy on Qwen3-VL and eight of ten lose significantly. The two exceptions, PruneVID (−0.35) and DyCoke (−0.67), are the two that retain roughly half their tokens against 7.5–15% for the rest (Appendix A).

Loss does not track budget monotonically. HoliTom operates at the smallest measured budget in the set (7.5%) and loses 2.19 points, while AIM at 15% loses 6.55 — more than four times as much on a comparable nominal budget. The methods that hold up are those that merge non-retained tokens into survivors rather than discarding them outright: FlashVID (pre-LLM merge plus an inner-LLM compression stage) still loses 3.16 points despite merging rather than discarding, so merging reduces but does not eliminate the cost of a 15% budget on this backbone.

2.5 Per-category change

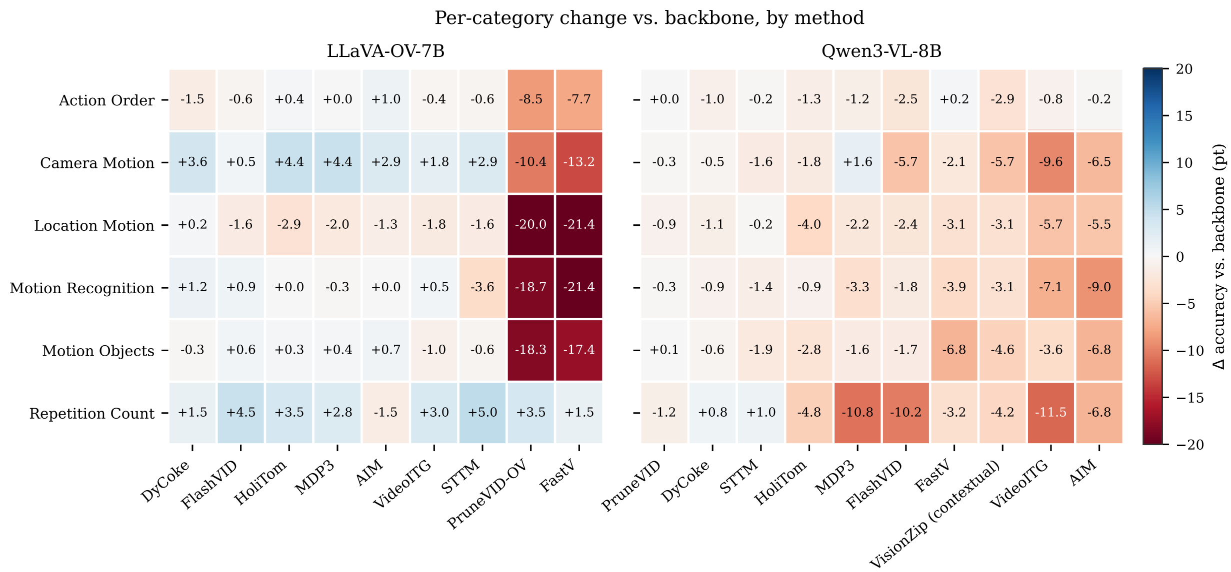

Figure 3: Per-category change vs. backbone, columns ordered by overall delta. Repetition Count movement is not interpretable — both backbones are at or near chance there (Table 2).

Losses are not spread evenly across question types. Expressing retained above-chance signal as (method − 25)/(backbone − 25), FastV on LLaVA-OV keeps **62%** of the backbone's above-chance signal on Motion-related Objects but only **30%** on Location-related Motion, 33% on Motion Recognition and 34% on Camera Motion. PruneVID-OV shows the same ordering (60% on Motion-related Objects, 34% on Location-related Motion). Motion-related Objects is the category most answerable from appearance in isolated frames; categories requiring integration across frames retain roughly half as much.

The pattern is milder but present on Qwen3-VL (Figure 3, right), where the largest losses fall on Motion Recognition (−9.0, AIM) and Camera Motion (−9.6, VideoITG), while Action Order moves within ±3 points for every method.

Repetition Count is the control column. LLaVA-OV scores 23.8% there against 25% chance, so there is no signal to preserve; the apparent gains under FastV (+1.5) and PruneVID-OV (+3.5) accompany 15-point overall collapses and reflect answer redistribution (§2.8), not recovered capability.

2.6 Retention sweep

Table 5: The three methods with an authors'-validated multi-point retention sweep, at every published point, ordered 10/15/20/25%. Baseline row spans all four columns since retention does not apply to it. * = significant vs. that backbone's baseline. — = not run: 20% was never run on Qwen3-VL for any method, and FastV's LLaVA-OV cells already span 10–75% (Table 8), making a 20% point redundant there. ‡ FlashVID's Qwen3-VL 10%/25% cells predate the authors'-code fix (§C): the incomplete reimplementation, not the verified method. Span is 10%→25% movement; not reported for FlashVID/Qwen3-VL since neither endpoint is the verified method. PruneVID is not in this table — its authors publish exactly one operating point, not a sweep (Table 6).

Method	$r = 0.10$	$r = 0.15$	$r = 0.20$	$r = 0.25$	Span
LLaVA-OV-7B					
Baseline		*52.66*			—
FastV	36.06*	36.78*	—	36.73*	+0.67
FlashVID	52.51	53.31	53.71*	53.36	+0.85
HoliTom	52.76	53.14	53.06	53.41	+0.65
Qwen3-VL-8B					
Baseline		*62.52*			—
FastV	56.92*	59.01*	—	60.60*	**+3.68**
FlashVID‡	54.73*	59.36*	—	59.36*	—
HoliTom	60.05*	60.33*	—	60.43*	+0.38

Table 6: PruneVID's true sweep. The retention tables in an earlier revision of this report labeled PruneVID's middle column "0.15"; that cell is `cluster_ratio=0.5`, the authors' only published point, and PruneVID was never run at 0.15 on either backbone. Sorted by the ratio that actually ran, Qwen3-VL is monotonic — the "non-monotonic" cell reported in an earlier revision was an artifact of the mislabeled column, not a property of the method.

Backbone	$c = 0.10$	$c = 0.25$	$c = 0.50$ (published)
LLaVA-OV-7B	38.10 (–14.56)*	38.38 (–14.29)*	38.20 (–14.46)*
Qwen3-VL-8B	60.23 (–2.29)*	61.40 (–1.12)*	62.17 (–0.35)

Tripling the budget from 10% to 25% moves no LLaVA-OV method more than 0.85 points. On Qwen3-VL the same change moves FastV 3.68 points, with its significance decaying monotonically ($\chi^2 = 89.8 \to 44.6 \to 18.2$) as it converges toward the baseline. FlashVID's equivalent Qwen3-VL span, 4.63 points, is not a claim about the verified method — it is computed entirely from the pre-fix reimplementation (Table 5, ‡ note); the verified 15% point (59.36%) does not sit on a verified curve at 10% or 25%.

On LLaVA-OV FastV and PruneVID sit roughly 15 points below the backbone at every setting tried, while FlashVID and HoliTom stay near the baseline. Within FastV's window, what costs accuracy is that it discards tokens at all, not how hard: the range was extended beyond the original 10–25% sweep to the authors' full published boundary, 12.5–75% keep, and the LLaVA-OV cells stay flat at 35.7–36.8% across all of it (Table 8, §2.7), including the authors' own K=2/R=50% operating point (36.34%). The window is bounded, however: §2.9 shows the same code path at 100% retention returns to the baseline, so the threshold sits strictly between 75% and 100% retention — narrower than previously reported, not located more precisely than that.

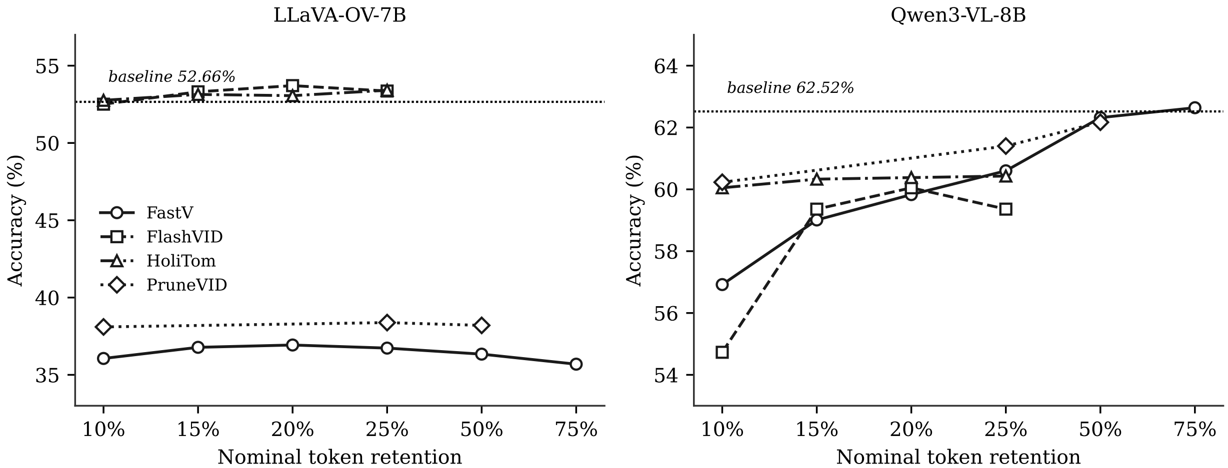

Figure 4: Accuracy against nominal token retention. On LLaVA-OV the four methods occupy two flat regimes the budget does not move; on Qwen3-VL the same methods respond strongly to the same parameter.

PruneVID's true sweep (Table 6) is monotonic on Qwen3-VL once sorted by the ratio that actually ran: $60.23 \to 61.40 \to 62.17$. An earlier revision of this report labeled the 0.5 cell "0.15" and reported the resulting order ($60.23 \to 62.17 \to 61.40$) as a flagged non-monotonic anomaly; that anomaly was an artifact of the mislabeled column, not a property of PruneVID.

FlashVID's 20% LLaVA-OV cell (Table 5) is the one retention point in this report that crosses significance in the direction of a gain: +1.05, $\chi^2 = 3.91$ against the 3.84 threshold (236 fixed vs. 194 broke). HoliTom's 20% cell does not (+0.40, $\chi^2 = 0.43$), consistent with its other three points. Both gate clean. The FlashVID result survives at only one of four points and the resolution floor is ± 1.54; it is reported because it crosses the threshold, not because a fourth data point changes the overall picture that FlashVID is accuracy-neutral on LLaVA-OV.

2.7 Which retention parameters are movable

Only three methods have a retention knob the authors themselves validated at more than one operating point — the bar for calling a value within that range "authors'-validated" rather than an extrapolation. Table 5 covers exactly those three; the remaining methods with any retention-like parameter fail this bar for the reasons in Table 7 below.

Table 7: Which methods' retention parameters are validated by the authors at more than one point, versus run here at a single point or not comparable at all.

Method	Authors' validated values	Status here
FlashVID	10/15/20/25% (identical on all four backbone scripts the authors ship)	run at all four, both backbones (Qwen3-VL 20% not run; see below)
HoliTom	10/15/20/25% (their own sweep script)	run at all four, LLaVA-OV; three, Qwen3-VL
FastV	12.5/25/50/75% keep	run at 10/15/25/50/75%, LLaVA-OV; three, Qwen3-VL
PruneVID	one point, `cluster_ratio=0.5`	run at that point plus two unpublished
DyCoke	no single ratio; one fixed recipe	run at the one published recipe
AIM	none on LLaVA-OV; unpublished, self-ported knob on Qwen3-VL	LLaVA-OV: fixed merge, no knob. Qwen3-VL: our own port, no authors' basis
STTM, VisionZip	per-benchmark configs or fixed token budgets, not a percentage	not a sweepable single percentage

FlashVID and HoliTom were run at their authors' full four-point LLaVA-OV sweep (Table 5); the 20% cell is discussed in §2.6 above.

FastV's authors publish a wider range (12.5–75% keep) than FlashVID or HoliTom's four points, which is why §2.6's threshold search above could extend to 50% and 75% without leaving validated territory:

Table 8: FastV's full LLaVA-OV-7B sweep (baseline 52.66%), extending Table 5 to the authors' full published boundary. K=2/R=50% is the authors' own most-published operating point.

Keep	10%	15%	25%	50% (K=2/R=50%)	75%
Overall	36.06	36.78	36.73	36.34	35.69
Δ	–16.60*	–15.88*	–15.93*	–16.33*	–16.97*

Every FastV LLaVA-OV cell between 10% and 75% is either an authors'-published point or an interpolation strictly inside their own tested boundary, never an extrapolation past it.

2.8 Answer distribution on LLaVA-OV

Table 9: Distribution of predicted answer letters on LLaVA-OV-7B, % of 4,018 scoreable items.

Run	A	B	C	D
Ground truth	*25.4*	*25.6*	*25.5*	*23.6*
Baseline	27.1	26.6	26.9	19.3
DyCoke	27.4	26.1	27.0	19.3
FlashVID	27.7	25.6	25.7	20.8
HoliTom	26.1	24.1	27.2	22.5
PruneVID-OV	12.2	26.7	15.0	**46.0**
FastV	14.9	23.0	15.2	**46.7**

The two collapsed methods place 46–47% of their answers on option D against a 23.6% ground-truth rate, while the backbone and the three merge-based methods stay near uniform. This is the behavior of a model answering from the prompt without usable visual evidence, and it explains the apparent Repetition Count gains in §2.5: a near-uniform category rewards a degenerate constant answer at roughly chance.

The signature is retention-invariant. Across $r \in \{0.10, 0.15, 0.25\}$ FastV's D-rate is 47.0%, 46.8%, 47.1% — flat to a tenth of a point while accuracy moves 0.67 points. Restoring tokens does not restore visual grounding on this backbone, though it does for the same algorithm on Qwen3-VL, where the answer distribution tracks the backbone within 1.5 points at every budget.

Why two rows, not one. FastV and PruneVID-OV are different algorithms at different layers with different selection rules, yet fail with signatures matching to about one point on accuracy, D-rate, broken/fixed ratio and flatness across retention. Two unrelated algorithms do not normally fail identically, and both LLaVA-OV ports drive reduction through the same code path — so the collapse could equally have been a property of that shared integration rather than of either method. §2.9 settles this by experiment.

2.9 What causes the collapse

Three explanations survived §2.8: the shared code path is defective; FastV's attention ranking is uninformative at video scale; or the budget concentrates on too few frames. A four-arm experiment separates them. Each arm holds the backbone, prompt, frame count, budget and decoding fixed and varies only the token-selection rule applied at the pruning layer, on a 986-question subset whose category mix matches the full benchmark within 1.5 points (Table 10).

Table 10: Selection-rule experiment on LLaVA-OV-7B, 15% budget except where noted, $n = 986$. *Broke:fixed* is discordant pairs against the backbone on the same subset.

Arm	Selection rule	Overall	D-rate	Broke:fixed
Baseline	*no reduction*	*54.36*	*20.7*	—
`keepall`	same code path, discards nothing	**55.07**	20.8	0.36
`attention`	FastV as published	39.25	49.4	2.27
`random`	uniformly at random	38.74	47.2	2.43
`uniform`	evenly spaced across all 32 frames	38.13	49.9	2.43

The code path is sound. `keepall` traverses the identical reduction machinery while discarding nothing and returns 55.07% against a 54.36% subset baseline, with a near-baseline D-rate and a broke:fixed ratio below 1. The shared integration is not the cause, and both LLaVA-OV rows stand as measurements of the methods.

The attention ranking is inoperative. `attention` and `random` differ by 0.51 points, $\chi^2 = 0.16$. FastV's layer-2 ranking performs no better than discarding 85% of visual tokens at random.

Frame coverage does not rescue it. `uniform` guarantees every frame contributes equally and is statistically indistinguishable from both other arms ($\chi^2 = 1.54$ and 0.30). Per-frame token histograms confirm the premise was wrong to begin with: every arm, `attention` included, touches all 32 frames on essentially every sample (mean 32.0 of 32), with the busiest frame taking 10.5% of `attention`'s budget against 3.1–4.2% for the others — far from concentration on a handful of frames.

What remains is a threshold effect. At a 15% budget on this backbone, discarding tokens costs about 15 points *regardless of which tokens survive*; at 100% the backbone is intact. Selection quality, and the temporal structure of the selection, stop mattering below that line. This is also the most economical account of why FastV and PruneVID-OV — unrelated in design — converge on nearly the same number: below the threshold there is nothing left for a selection rule to differentiate. The threshold itself was subsequently located more precisely (§2.6): FastV stays flat from 10% to 75% keep, including the authors' own K=2/R=50% setting, and recovers only at 100%, so the transition sits strictly between 75% and 100%, not measured more finely than that.

PruneVID-OV corroborates the same account independently. Setting `cluster_ratio=1.0` disables its spatial merge — the analogue of `keepall` for a structurally different algorithm at a different layer — and scores 53.36%, +0.70 against the backbone, $\chi^2 = 2.34$, not significant: the same recovery FastV's own `keepall` control shows. Two unrelated methods, discarding through unrelated mechanisms, both intact with nothing discarded and both collapsed the moment anything is — this is a property of the backbone below some retained-token density, not of either method's selection rule.

The fork used here reproduces the production FastV run exactly on the matched subset (0 of 986 predictions differ), so the experiment measures the same object as Table 3.

2.10 Methods on non-standard backbones

Table 11: Runs on backbones other than the two standardized ones. Not comparable to Tables 3–4; no Δ is given because no matched baseline exists for either row.

Method	Backbone	Overall	AO	CM	LM	MR	MO	RC
VisionZip	LLaVA-1.5-7B	40.09	33.7	33.0	37.2	41.1	57.4	25.8
DyTo (recon. TW-FINCH)	Vicuna-7B	42.06	32.4	33.0	42.1	43.0	61.7	26.0

VisionZip patches `CLIPVisionTower`, which LLaVA-OV does not have, so its complete form runs only on LLaVA-1.5-7B — and at 8 frames rather than 32. The 40.09% differs from Table 3 in both backbone and frame count and is never placed in a LLaVA-OV comparison.

DyTo carries two caveats. It is not reproducible from its published artifacts (the released code calls an argument absent from its own pinned dependency, and a clustering helper has no return statement); the missing return was recovered by transcription from a character-identical sibling function, but the temporal weighting had to be reimplemented, so the result is reported only as "DyTo (reconstructed TW-FINCH)". And it is the one number in this report with execution evidence but **no divergence check**, because no baseline was ever run on its Vicuna backbone — it is an absolute score, not a measured method effect.

A token-matched control is documented in the project records (cited, not recomputed here): against an uncompressed 6-frame baseline at 3,456 visual tokens scoring 43.35%, DyTo at ~3,680 tokens scores 42.06%, a significant −1.29. A frame-matched control is impossible — Vicuna's 4,096-token context cannot hold 32 uncompressed frames — which is the constraint DyTo exists to address.

A Methodology and verification

The verification gate

A run that completes, reports a plausible accuracy, and did nothing is indistinguishable from a successful one by accuracy alone. Every number in this report therefore required two independent signals.

1. Execution evidence. The method's instrumentation reported a token count consistent with its configured budget (Appendix A).

2. Prediction divergence from the unmodified backbone. Greedy decoding makes the pipeline a deterministic function of (weights, frames, prompt): frame indices come from a fixed linear spacing, the vision tower is frozen, token choice is $\arg\max$. Re-running an unchanged configuration reproduces all 8,052 predictions exactly — confirmed on three independent run pairs, each diverging on 0 of 8,052. Against that zero baseline, a method that discards or merges most of the visual input and produces zero differing predictions cannot have executed.

The second signal is not redundant. Four ports passed every other check and failed it: an early FastV integration whose pruning call was a stub; a PruneVID port to LLaVA-OV whose discard mask was dropped before reaching the model; and FlashVID and AIM on Qwen3-VL, which logged a correct budget (`keep=1750 (15.0%)`) while emitting byte-identical output, because `attention_mask` is `None` under this model's default attention implementation and the mask-dependent path never ran. A fifth, STTM on Qwen3-VL, failed because that model interleaves timestamp text tokens between frames — the 11,664 visual tokens occupy an 11,784-wide span broken by 15 gaps, and a patch assuming one contiguous block did nothing. All five would have entered the results tables as working methods under an execution-log check alone.

Measured token budgets

Table 12: Measured visual-token budget on Qwen3-VL-8B, read from each method's execution log rather than its CLI flag. Input is 11,664 visual tokens at 32 frames. Nominal 15% describes five of ten methods.

Method	Reduction stage	Measured budget
HoliTom	pre-LLM segment retain + in-LLM merge	7.5% (875 tokens)
STTM	quadtree spatio-temporal merge, published setting	not re-measured since the off-spec→published fix
FastV	in-LLM attention rerank, layer 2	15.0% (1,750)
FlashVID	pre-LLM merge + inner-LLM compression, layer 28	15.0% (1,750) pre-LLM; further compressed mid-network, not captured here
AIM	bipartite merge + PageRank prune	15.0% (1,750)
VisionZip (contextual)	key-similarity merge in vision tower	15.0% (1,750)
DyCoke	temporal merge + KV prune, layer 3	49.0% (5,716)
PruneVID	DPC-KNN cluster merge, layer 10	50.0% (5,832)
MDP3	frame selection (DPP)	8 of 32 frames
VideoITG	grounded frame selection	32 of 512 sampled

Provenance

Every figure in the results tables and all four figures were recomputed for this report directly from each run's per-sample prediction records, not transcribed from prior documentation. Recomputed accuracy agreed with each run's stored summary to within 0.01 points across all available run directories, and independently reproduced the documented divergence counts, paired win/loss counts and per-category figures. The DyTo token-matched control (§2.10) is cited rather than recomputed, as no baseline run exists on its backbone.

Six Repetition Count cells land on an exact rounding tie: HoliTom and PruneVID-OV at 109/400 = 27.25%, DyCoke and FastV at 101/400 = 25.25%, AIM at 89/400 = 22.25% and STTM at 115/400 = 28.75% (all LLaVA-OV). A seventh, introduced by the Qwen3-VL STTM correction (§C), is 139/400 = 34.75%. This report rounds half away from zero throughout, giving 27.3, 25.3, 22.3, 28.8 and 34.8. Python's default banker's rounding disagrees on every one of them, so recomputation scripts must not be used to "correct" these cells without changing the stated convention first.

B Paired-prediction statistics

Differ counts predictions unequal to the backbone's over all 8,052 rows (accuracy uses the 4,018 scoreable subset); 0 would indicate a silent no-op. *Fixed* and *broke* are the McNemar discordant pairs on scoreable items.

Divergence volume does not predict damage, and the two must not be conflated. STTM on LLaVA-OV changes 4,859 predictions — more than FastV's 4,605 — yet loses 0.94 points and is not significant, because its changes fall almost evenly in both directions (329 fixed, 367 broken). FastV changes a comparable number destructively (475 fixed, 1,113 broken). Divergence establishes only that a method executed; the discordant pairs establish what it cost.

Table 13: Paired-prediction statistics against each backbone.

Method	Differ	Fixed	Broke	χ^2	Significant
LLaVA-OV-7B					
DyCoke	1,031	170	142	2.34	no
FlashVID	1,610	268	242	1.23	no
HoliTom	1,838	299	280	0.56	no
MDP3	1,452	246	230	0.47	no
VideoITG	1,193	192	184	0.13	no
AIM	1,406	227	219	0.11	no
STTM	4,859	329	367	1.97	no
PruneVID-OV	4,536	473	1,054	220.30	yes
FastV	4,605	475	1,113	255.52	yes
Qwen3-VL-8B					
PruneVID	1,074	56	70	1.34	no
DyCoke	1,048	83	110	3.50	no
HoliTom	1,657	131	219	21.63	yes
MDP3	1,626	159	274	30.01	yes
FastV	2,029	149	290	44.65	yes
VisionZip (contextual)	2,035	191	340	41.25	yes
STTM	1,510	133	170	4.28	yes
FlashVID	2,069	157	284	36.00	yes
VideoITG	2,522	206	454	92.44	yes
AIM	2,861	194	457	105.44	yes
Backbone vs. backbone					
Qwen3-VL-8B vs. LLaVA-OV-7B	7,081	810	414	127.47	yes

C Limitations

Scope. All results are on MotionBench and two backbones. Nothing here generalizes to other video benchmarks; the per-category analysis of §2.5 in fact predicts that these methods will look better on benchmarks weighted toward appearance and object recognition.

Single measurement per cell. Greedy decoding makes each configuration exactly reproducible, so there is one measurement per cell and no run-to-run variance estimate. Reported uncertainty is binomial, not empirical — which is why the ±1.54-point floor is applied strictly.

Standardization is not paper fidelity. The headline comparison tables (Tables 3, 4) fix every method to a common 15% nominal budget for cross-method comparability, which is not each method's own published default. FastV's LLaVA-OV cells were additionally run across its full published range, 10–75% keep (§2.6), including its own K=2/R=50% operating point, so that specific comparison is no longer standardization-only; PruneVID's headline cells run at its published `cluster_ratio=0.5`, not 15% (Table 1 † note). Since methods without a retention knob keep their defaults, the cross-method comparison is at a common nominal setting and not at equal compute (Appendix A).

Determinism is conditional. Bit-identical reproduction was verified across nodes of the same GPU generation, same software stack, batch size 1. Different GPU generations may perturb near-tied logits; that comparison was not run. Replicators on other hardware should expect close, not identical, numbers.

Zero divergence is strong but not airtight. A method pruning only tokens the model already ignored would execute correctly and produce zero divergence, and the gate would misclassify it. No such case was observed.

Not every cell runs the authors' code. The gate establishes that a mechanism ran and changed the output, never that it matches the published algorithm. On LLaVA-OV most cells execute upstream code; on Qwen3-VL, which postdates most of these methods, most are our ports and should be read as "our implementation of the method". Two cells previously flagged here have since been resolved: FlashVID on Qwen3-VL now runs the authors' own `flashvid()` package with inner-LLM compression at its published setting (59.36%, replacing a 56.65% reimplementation cell that omitted that stage), and STTM on Qwen3-VL now runs a published configuration (61.60%, replacing an off-spec `root_level=0` cell at 57.07%). One cell remains genuinely unresolvable: PruneVID on LLaVA-OV has no upstream reference at all — its released code targets PLLaVA only, so the 38.20% cannot be validated while its PLLaVA result (44.13%, Table 11) can. AIM's Qwen3-VL retention knob is a further case worth naming explicitly: the authors ship no Qwen3-VL code at all, so `prune_ratio` on that backbone is entirely our own port, with no authors' configuration to validate against at any value (§2.7). Per-cell provenance is recorded in the project repository.

Scoring protocol. Accuracy depends on a letter-matching regular expression and on excluding the 4,034 `NA` items — roughly half the benchmark. Both are documented choices; neither has been ablated.

D Open items

Three items from an earlier revision of this report are now closed; they are recorded here for continuity rather than left silently to disappear.

Closed. Where the discard threshold falls. FastV's LLaVA-OV cells were extended from the original 10–25% sweep to 10/15/25/50/75% keep, covering the authors' full published range and their own K=2/R=50% operating point. All five land at 35.7–36.8%; recovery happens only at 100% (§2.9, Table 8). The threshold is narrower than previously stated — strictly between 75% and 100% — but was not located more finely than that.

Closed. FastV at its stated video setting. K=2/R=50% (36.34%) and K=2/R=75% (35.69%) were both run; both collapse in the same band as the standardized K=2/R=85% (keep 15%) cell. FastV's published operating point does not recover the LLaVA-OV collapse.

Closed. Re-runs at published configurations. All three pending re-runs landed: FlashVID on Qwen3-VL now runs the authors' `flashvid()` package (59.36%, replacing 56.65%); STTM on Qwen3-VL now runs a published setting (61.60%, replacing an off-spec 57.07%); PruneVID-OV's `cluster_ratio=1.0` control recovers to 53.36%, corroborating FastV's own `keepall` result independently (§2.9).

Still open. Whether the threshold is backbone-specific. No four-arm selection experiment analogous to §2.9 was run on Qwen3-VL. The retention sweep (§2.6) shows the same methods respond smoothly to budget there with no collapse of any size, which is suggestive but not the same causal test run on LLaVA-OV.

New. FlashVID's Qwen3-VL 10%/25% cells are stale. Only the 15% cell has been re-run against the authors' code; the 10% and 25% cells in Table 5 still reflect the incomplete reimplementation. Re-running those two at the authors' settings would let FlashVID's Qwen3-VL retention sensitivity be reported on verified data throughout, the way FastV's already is.